

Daniel Gordon

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EDUCATION

Honours Computer Science and Management (B.Sci), Wilfrid Laurier University

September 2022 - Dec 2026

Research Interests: Deep Learning, Reinforcement Learning, Time-Series Forecasting, Optimization, NLP, Evolutionary Computation

Waterloo, ON

MACHINE LEARNING

Research Assistant – AI/ML Research Project

September 2025 - Present

Wilfrid Laurier University

Waterloo, ON

- Verified Sarukhanian's δ -code constructions by implementing computational checks that identified a sign error in Proposition 1; corrected the formula and produced a zero-NPAF sequence matching theoretical expectations.
- Applied evolutionary and stochastic optimization (Genetic Algorithms, Simulated Annealing) to repair the incorrect Proposition 2 structure, reducing NPAF errors from **16 non-zero shifts to 3** across a search space $>10^{20}$ combinations.
- Combined exhaustive search (2^{16} sign patterns, 16k sequence combinations) with symbolic Maple verification to mathematically disprove the published construction and validate all computational findings.

N-Queens Reinforcement Learning Solver (Python)

January 2025 - April 2025

Wilfrid Laurier University

Waterloo, ON

- Implemented an optimized **Min-Conflicts** CSP solver capable of scaling from $N=10$ to $N=1,000,000$, achieving solution times of ~ 20 seconds on standard hardware.
- Designed greedy initialization, constant-time conflict repair, and inverse-index conflict tracking to reduce runtime from $O(n^2)$ to $O(n)$.
- Built benchmarking tools and validation utilities; ran large-scale experiments to verify correctness and measure performance across increasing N .

ML Tools: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, statsmodels, supervised learning, RL, sequence modeling, error analysis, benchmarking, ablations Genetic Algorithms, Simulated Annealing, Exhaustive Search

EXPERIENCE

Contract Full Stack Developer

April 2024 - Present

Self-Employed

Vaughan, ON

- Built an AI-driven WhatsApp summarization platform using the WhatsApp Web API + **Gemini LLM**, serving ~ 200 daily users and processing $>1M$ tokens/day of unstructured text.
- Designed end-to-end NLP pipelines for ingestion, batching, cleanup, summarization, and hallucination mitigation through iterative prompting.
- Optimized latency and context-window usage via dynamic chunking + caching strategies inspired by long-context LLM techniques, reducing average response time by $\sim 30\%$.
- Developed modular REST APIs and React/Next.js interfaces used by 5+ clients. Delivered complete project lifecycles with consistent delivery.
- Implemented monitoring and logging tooling to analyze summarization failures, user behaviour, and model efficiency, informing iterative improvements to the NLP pipeline.

Full Stack / AI Developer

July 2025 - Present

MCG - Digital Assetization Company

Vaughan, ON

- Refactored a Python/JavaScript lip-sync ML pipeline for 3D avatars in Unity, improving phoneme-to-blendshape alignment and enabling both audio- and text-driven animation triggers.
- Prototyped end-to-end AI workflows: build Whisper-based speech-to-text and LangChain orchestration, benchmark alternatives, and recommend the best accuracy/latency configuration.
- Implemented model-level optimizations (quantization, beam-search tuning, batching), reducing inference latency $\sim 25\%$ without degrading accuracy.
- Expose lip-sync and AI capabilities as RESTful services and integrate with a React/Next.js front end to drive real-time avatar animations in the browser.

Junior Solution Architect and Developer

June 2025 - August 2025

LaunchPath Inc.

Vaughan, ON

- Architected a scalable system for a biosensor analytics platform, producing C4 diagrams, technical specifications, and end-to-end integration plans.
- Led Agile delivery across 3 sprints, defining epics/user stories and presenting sprint demos to stakeholders.
- Built a responsive React Native interface with secure integration to Python microservices for real-time biosensor data ingestion.

PROJECTS

- **TourSage – AI Tour Guide App:** Integrated generative AI + RAG (semantic search, embeddings) for itinerary generation using Python backends and Node/TypeScript.

- **wluNest Housing App:** Implemented search + geospatial queries.

- **Battleships AI Game (Java):** Implemented heuristic search with minimax-style decision trees and pruning heuristics.

SKILLS

Programming Languages: Python, Java, JavaScript, C/C++, Lua, TypeScript, Dart, HTML, CSS, SQL, R, Bash, Maple

Technologies & Tools: Git, Microsoft Office, React, React Native, Next.js, Flutter, VS, Apache Maven, Prisma, Tailwind, WireShark, MySQL, MongoDB, PostgreSQL, Figma, Framer, [Node.js](#), Electron, [Babylon.js](#), Flask, Unix, Expo, Unity, Hugging Face Transformers, Venv, Shell Scripting

Mathematical Foundation: Linear Algebra (eigenvalues, SVD, PCA), Probability (Bayes, MLE), Optimization (gradient descent, convexity), Information Theory (entropy, KL divergence)